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Inverse Problems for Advanced Nozzle Design

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The aerodynamic design of classical and innovative nozzle concepts is investigated by an inverse method. The proposed numerical technique solves two-dimensional/axysimmetric viscous inverse problems that arise in aerodynamic design. The approach, which is well established for inviscid, rotational flows, is here extended to the viscous case. The methodology is coupled to an optimization technique seeking for wall pressure gradient distributions that maximize the nozzle thrust.

Nomenclature

D	= domain in $(x, y) \in R^2$
∂D	= boundary of D
$\boldsymbol{i}, \boldsymbol{j}$	= unit vectors
$\boldsymbol{F}_I$	= tensor of convective fluxes
$\boldsymbol{F}_V$	= tensor of diffusive fluxes
$\boldsymbol{m}$	= wall pressure gradient
$\boldsymbol{n}$	= normal unit vector
$\boldsymbol{q}$	= velocity vector
$\boldsymbol{W}$	= vector of conservative variables
u, v	= Cartesian velocity components
Subscript	
in	= inlet
ou	= outlet

I. introduction

The reduction of Earth-to-Orbit launch costs in conjuction with an increase in launcher reliability and operational efficiency are the key demands on future space transportation systems, like Single-Stage-to-Orbit vehicles. The realization of these vehicles strongly depends on the performance of the engines, which should deliver a high performance at low system complexity. Performance data of rocket engines are practically always lower than the theoretically attainable values due to imperfection in mixing, combustion and expansion of the propellants. In particular, a relevant source of thrust loss (up to 15%) is related to non-adapted flow conditions during the ascent. The attention is then addressed to nozzle concepts, which promise, at least theoretically, benefits in performance over conventional nozzles, achieved by altitude adaptation, and thus by minimizing losses caused by over- or under-expansions. Experimental, analytical and numerical research on plug nozzle has been performed since the 50's in several countries, see e.g. (1,2) In contrast to the classical

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nozzle concepts plug nozzles provide, at least theoretically, a continuous altitude adaptation up to their geometrical area ratio. Two different design approaches for circular plug nozzles are possible: plug nozzle with a toroidal chamber and throat (with or without truncation), and plug nozzles with a cluster of circular bell nozzle modules or with a cluster of rectangular nozzle modules. A further plug nozzle configuration is the linear plug nozzle, which is foreseen for the propulsion system of the RLV X-33 concept. The interest for these nozzles relies on their self-adaptable characteristics, that can positively circumvent one of the limits of conventional nozzles. Indeed, to face the decrease of ambient pressure along the mission trajectory, the conventional nozzle design is performed on the basis of a compromise between the requirement of large expansion ratio, to have better performance at high altitudes, and the limitation in the over-expansion at sea level, to avoid uncontrolled flow separation and consequent side loads. Another, adaptive nozzle concept is the dual-bell nozzle, first proposed by Forster and Cowles[3] in 1949. The dual-bell nozzle has gained renew interest for its unique feature of an one-step altitude adaptation, achieved only by a wall inflection and, thus, without moving parts. Because of the classical bell-shaped form, necessary changes on the launcher level are small when compared to other altitude adaptive nozzle concepts, for example plug nozzle. At low altitudes, controlled and symmetrical flow separation occurs at this wall inflection, which results in a smaller effective area ratio without generating dangerous side loads. At higher altitudes, the nozzle flow is attached to the wall until the exit plane and the full ratio is used. Because of the higher area ratio, an increase in vacuum performance is achieved. The first published experiments on dual bell nozzle were performed by Horn and Fisher.[4] Their cold-flow subscale nozzle tests proved the altitude adaptive capability of this nozzle, but also showed that additional losses are induced compared to a conventional bell-shape design. In 1999, Frey and Hagemann[5] presented results of analytical and numerical studies. The dependence of additional losses induced in dual-bell nozzles as a function of nozzle designed parameters was systematically investigated, and performance characteristics were compared to a conventional reference bell nozzle. It was shown that additional losses induced by the wall inflection depend on the contour design and can be surprisingly low.

In the present paper a technique of inverse problem solution is developed for designing conventional or unconventional nozzles. The methodology reformulates the design problem as an inverse problem, solved numerically using the compressible Navier-Stokes equations in two dimensions or for axisymmetric configurations. The inverse problem consists in finding the shape that the walls of a body have to assume to satisfy some flow properties prescribed over them as design data, the wall pressure in our case. The numerical method is based on a time-dependent integration of Navier-Stokes equations discretized using a finite volume approach. Detail are given in the first sections, while numerical experiments of designing linear plug and dual-bell nozzles are shown and the results are compared to experimental data from literature. A gradient-based optimization technique is used to solve the optimization constrained problem and to drive the inverse methodology towards the design of nozzle for maximum thrust. As a numerical experiment the optimization of a bell nozzle for maximum thrust is shown, being fixed the inlet and outlet pressure levels and according to a constraint criterion to avoid uncontrolled flow separations.

II. Governing Equations

Compressible viscous flows are governed by the Navier-Stokes equations. This set of equations may be written in a compact integral conservative form as:

$$\frac{\partial}{\partial t}\int_{\mathcal{V}} \boldsymbol{W}\, d\mathcal{V} + \int_{\mathcal{S}} (\boldsymbol{F}_I - \boldsymbol{F}_V)\cdot \boldsymbol{n}\, d\mathcal{S} = 0 \tag{1}$$

where $\mathcal{V}$ represents an arbitrary volume enclosed in a surface $\mathcal{S}$. System (1) is reduced to non-dimensional form with the help of the following reference values: L for length, ρ_∞ for density, T_∞ for temperature, $\sqrt{RT_\infty}$ for velocity, RT_∞ for energy per unit mass and μ_∞ for viscosity. In particular, $\boldsymbol{W}$ is the vector of

conservative variables while tensors $\boldsymbol{F}_I$, $\boldsymbol{F}_V$ contain the inviscid and the viscous fluxes, respectively.

$$\begin{aligned} \boldsymbol{W} &= \{\rho, \rho\,\boldsymbol{q}, E\}^T \\ \boldsymbol{F}_I &= \{\rho\,\boldsymbol{q}, p\,\boldsymbol{I} + \rho\,\boldsymbol{q}\otimes\boldsymbol{q}, (E+p)\,\boldsymbol{q}\}^T \\ \boldsymbol{F}_V &= \frac{\sqrt{\gamma M_\infty}}{\mathrm{Re}_\infty}\{0, \boldsymbol{\tau}, \kappa\nabla T + \boldsymbol{\tau}\cdot\boldsymbol{q}\}^T \end{aligned} \tag{2}$$

The viscosity μ is computed via Sutherland's law and the perfect gas relationship $p = \rho T$ completes the set of equations.
A two-dimensional/axisymmetric formulation of system (1) is solved using a finite volume technique[6] on time-varying computational domains. The integration in time is carried out according to a two-step Godunov scheme. At the *predictor* step, a standard first order Flux Difference Splitting (FDS) scheme is used: the conservative variables $\boldsymbol{W}$ are assumed as an averaged, constant value inside each cell. The fluxes $\boldsymbol{F}_I$ are evaluated by solving the Riemann's problems pertinent to the discontinuities that take place at the interfaces of the cells. At the *corrector* level, the second order of accuracy is achieved by assuming a linear, instead of constant, behaviour of the conservative variables inside the cells, according to an Essentially Non-Oscillatory (ENO) scheme.[7] The resulting scheme is second order accurate in both time and space. The velocity and temperature gradients that are required to evaluate the viscous fluxes $\boldsymbol{F}_V$ in correspondence to the lateral surface are computed using a integral technique that utilizes central differences and applies Gauss's Theorem.

III. Boundary Conditions

The computational domain is bounded by artificial (i.e. far field boundaries) and physical contours (i.e. impermeable walls), which can be solid, as in the direct problem, flexible, as in the inverse problem, or partly solid and partly flexible. The boundary condition (BC) enforcement follows the guidelines given in Poinsot and Lele.[8] To the conditions corresponding to inviscid flows, supplementary relations, the *viscous conditions*, are added, to complete the set of boundary conditions needed by Navier-Stokes equations. The unconventional part of the boundary condition enforcement is the moving wall BC which is therefore discussed in next section.

A. Moving Wall Boundary

When solving the inverse problem, walls on which a flow variable, e.g. the pressure, is imposed, are treated as flexible and impermeable surfaces whose motion is tracked in time until a steady state is reached. In inviscid flows, the velocity vector must be tangent to the wall surface, which is therefore a flow surface and, *it will move materially within the fluid*. These considerations are still valid for viscous flows, but the set of boundary conditions must be completed with the additional viscous relations.
Let us consider a Cartesian coordinate system (x, y). A generic surface moving in time can be represented, for example, by equation $y = \beta(x, t)$. Without any loss of generality, let us also assume that $\beta(x, 0) = 0$. Thanks to the change of variable $\mathcal{B} = y - \beta$ the surface is now defined by the manifold

$$\mathcal{B}(x, y, t) = 0 \tag{3}$$

This manifold moves materially within the fluid, that is,

$$\frac{\partial \mathcal{B}}{\partial t} + \boldsymbol{q}\cdot\nabla\mathcal{B} = 0 \tag{4}$$

or

$$\frac{\partial \beta}{\partial t} = v - u\frac{\partial \beta}{\partial x} \tag{5}$$

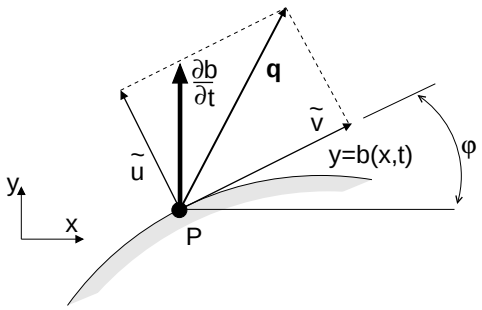


Figure 1. Inverse problem: moving wall velocity.

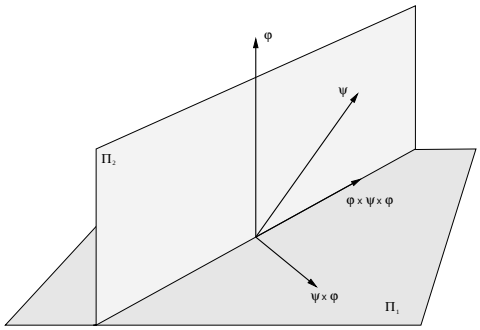


Figure 2. Projection of the gradient in the constraint space.

The flow velocity $\boldsymbol{q}$ at boundary is obtained by imposing the target pressure on the moving wall and solving the related Riemann problem. Once $\boldsymbol{q}$ is known at the boundary, from (5) the wall velocity $\partial\beta/\partial t$ can be deduced. The latter is then integrated in time to find the new wall shape. In this context, the no-through-flow condition represents a kinematical constraint from which the motion of the wall surface is derived. The grid is adjusted to fit the updated wall geometry at each time step. From geometrical considerations, as shown in Fig.1, an alternative form of the time derivatives $\partial b/\partial t$ is

$$\frac{\partial b}{\partial t} = \frac{\widetilde{u}}{\cos\varphi} = \widetilde{u}\sqrt{1+\left(\frac{\partial b}{\partial x}\right)^2} \tag{6}$$

The same inverse method can be used to determine the shape of plumes and interfaces. For instance, in flow regions such as after-bodies or dual nozzles in a by-pass turbofan, contact surfaces are generated by the different stagnation conditions and thermodynamic properties of the incoming flows. These discontinuities are interfaces that can be computed explicitly according to the present inverse method. In the inviscid case they are considered as impermeable and deformable boundaries separating different flow regions,[9] across which the pressure and the normal component of the flow velocity are imposed to be C^0 continuous and equal to the deformable boundary velocity.
As for the fixed wall boundary condition, the extension to the viscous case is obtained by adding the appropriate physical boundary conditions. For isothermal walls, suitable BC are

$$\widetilde{v} = 0, \qquad T = T_w \tag{7}$$

while for adiabatic walls or for the fitting of the previously mentioned contact surfaces, we impose

$$\widetilde{v} = 0, \qquad \frac{\partial T}{\partial n} = 0 \tag{8}$$

IV. Optimization

The nozzle has imposed inlet p_{in} and outlet p_{ou} wall pressures. The distributed control is the wall pressure gradient m. On the discrete level, the pressure is recovered as

$$p_w(x_i) = p_{in} + \sum_{j=2}^{i} m(x_j)\Delta x_j, \qquad m(x_j) = \left(\frac{dp_w}{dx}\right)_j \tag{9}$$

with the constraint

$$\sum_{j=2}^{N} m(x_j)\,\Delta x_j = p_{ex} - p_{in} \tag{10}$$

to match the exit pressure. Δx_j is the grid size in the x-direction and N is the number of computational points in the x-direction. We have $N-1$ control parameters represented by the pressure gradient at the discretization points. The global variation of thrust is

$$\delta T_{p_w} = \sum_{j=2}^{N} \frac{\partial T}{\partial m_j}\delta m_j = \sum_{j=2}^{N} \psi_j \delta m_j, \qquad \psi_j = \frac{\partial T}{\partial m_j} \tag{11}$$

If we take $\delta m_j = \psi_j$, then $\delta T > 0$ and the maximum is eventually reached. However this assumption does not guarantee that eq. (10) is satysfied and and the final pressure level p_{ou} does not remain constant. Yet, in view of eq. 10, we also obtain

$$\delta p_{ou} = \sum_{i=2}^{N} \varphi_i\, \delta m(x_i)\, \Delta x_i = 0 \tag{12}$$

If $0 \le m(x_i) \le \Pi(x_i)$, then $\varphi_i = 1$, otherwise we take $\delta m(x_i) = 0$ and $\varphi_i = 0$. $\Pi(x_i)$ is a given constraint on the pressure gradient, e.g. a criterion to avoid flow separations. By projecting the gradient on the plane tangent to the constraints, see fig. 2, we have $\delta T > 0$ by taking

$$\{\varphi_i\} \times \{\psi_i\} \times \{\varphi_i\} = \delta m(x_i) = \varrho \left(\psi_i \sum_{j=2}^{N} \varphi_j^2 - \varphi_i \sum_{j=2}^{N} \psi_j \varphi_j \right) \tag{13}$$

The solution of the optimization problem is achieved by initializing the coefficients $m(x_j)$, computing the corresponding wall geometry using the inverse problem, evaluating numerically the functional gradient and updating the coefficients $m(x_j)$, according to the projected gradient, until the maximum is reached. The gradient evaluation involves the additional solution of N inverse problems. Even if in this step the flowfield configurations are close each other, the computational cost is high, since N is about the number of nodes in the axial direction.

A. Flow Constraint

A simple form of empirical separation criteria is based on the Schilling[?] curve fit

$$\frac{p_{amb} - p_{sep}}{p_c} = C \left(\frac{p_c}{p_{amb}} \right)^E \tag{14}$$

where, for short contoured nozzle, $C = 0.517$ and $E = -0.913$. From Eq.(14) a constraint for separation control can be obtained as

$$p_w^* \ge p_{amb} - p_c C \left(\frac{p_c}{p_{amb}} \right)^E \tag{15}$$

where $p_w^* = p_{sep}$ is the wall pressure level at which separation occurs. To satisfy this constraint we set the local pressure gradient at x_j to a fraction of the maximum allowable gradient as

$$\Pi(x_j) = \left(\frac{dp}{dx} \right)_{j\ max} = \xi \frac{p_w^* - p_{w_{j-1}}}{x_j - x_{j-1}} \tag{16}$$

where $0 < \xi < 1$ is a constant (we used $\xi = 0.6$). This is just a shortcut for including separation on the design process. The reliability of modern flow solver allows moreover to predict the laminar and turbulent flow separation with good accuracy, so that the maximum wall pressure gradient can be evaluated during the automated loop of optimization that

1. sets an initial wall pressure distribution $p_w(x)$ and maximum allowable wall pressure gradient $m^*(x)$
2. solves the inverse problem
3. evaluates numerically the flow smoothness close to the wall and detects an incipient separation
4. updates the $m^*(x)$ distribution to safer values near the separation point
5. evaluates the functional gradient
6. marchs as stated by (13) to satisfy the constraint
7. goes to point 2 until convergence is reached

V. Numerical Results

With the aim of validating the viscous inverse solver and of focusing on its accuracy, some numerical experiments are presented. These tests illustrate how the nozzle design can be naturally approached by inverse methods. When possible, the computed flowfieds and shapes are compared with theoretical solutions or available experimental data. The final example refers to a possible way of coupling of the method with an optimization technique.

A. Linear Plug Nozzle

Following Angelino's method,[10] an ideal contour of the plug has been defined, which will hereafter be called the *target plug contour*. We set the design pressure ratio $PR = 200$. The Reynolds number, based on the plug length, was set at $Re = 10^7$, a value which is in the range of practical applications. Regardless of the Reynolds number, the flow is assumed to be laminar. To set up the test case, for the given plug geometry, the wall pressure is computed using a direct Navier-Stokes flow solver. The obtained pressure distribution is then enforced as the target pressure at the plug wall and the inverse problem is solved for the nozzle geometry. No matter what the initial condition is, we expect the final geometry to accurately replicate the target plug contour. The numerical test was solved on a 100×52 grid. Stretching was used in the y-direction, to better refine the region close to the upper and lower boundaries, and also in the x-direction, to cluster points in the region where most of the expansion takes place. The upper boundary is implemented as a far-field condition, while the bottom boundary moves according to the procedure of inverse designing. A sequence of the flowfield evolution during the transient is presented in Figure 3. The starting geometry is visible in Figure 3a. The target flow solution is characterized by a continuous expansion, without discontinuities, while the presence of a shock and its reflection are quite visible in the initial flowfield. The initial flowfield can therefore be considered sufficiently far from the target one. The inverse problem procedure is driven by flow evolution, so that the complexity of the initial flowfield, rather than the initial geometry of the domain, is of relevance. A typical evolution of the boundary subjected to conditions (6) and (7) or (8) dictated by the inverse procedure is also visible in Figure 3. After a motion that somehow resembles a "whip lash", the contour assesses to the imposed boundary pressure, and a microscopic motion, on the scale of the boundary-layer thickness, takes place to balance the viscous fluxes in the proximity of the wall. The plug shape is obtained with high accuracy. An expected result is the good agreement between the imposed and the obtained wall pressure distribution, as confirmed by the grid refinement study in Figure 4. The computed geometries and wall pressure distributions are obtained on three different grid sizes 50×30, 100×52, 150×80, respectively. The results confirm that the methodology is independent of the grid size and a logarithmic scaling is necessary in Figure 4 to highlight minimal discrepancies between the target and computed plug contours. The highest nismatch is observed in the proximity of the spike end, the relative error in the plug geometry being of the order of 10^{-3} for the coarsest grid (50×30 cells).

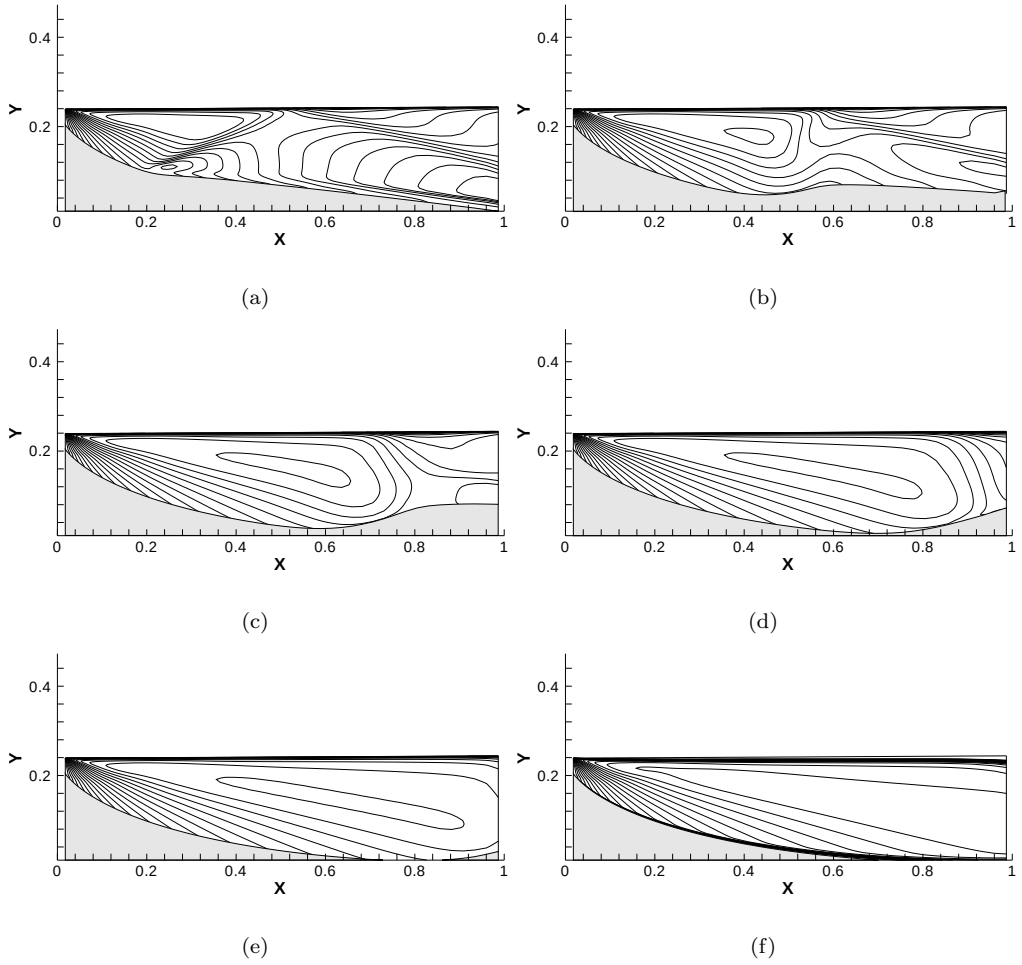


Figure 3. Inverse problem on a plug nozzle. Snapshots of the evolving flowfield in terms of Mach number isocontours. (a) initial, (b,c,d,e) intermediate, (f) final flowfield .

B. Dual Bell Nozzle

The example refers to the design of an axysimmetric dual bell nozzle. This nozzle concept is of actual interest for its unique feature of a one-step altitude adaptation, achieved only by a wall inflection and, thus, without moving parts. At low altitudes, controlled and symmetrical flow separation occurs at the wall inflection, which results in a smaller effective area ratio without generating dangerous side loads. At higher altitudes, the nozzle flow is attached to the wall until the exit plane and the full ratio is used. Because of the higher area ratio, an increase in vacuum performance is achieved. The overall design parameters are kept from Hagemann et al.[11] and refer to the truncated ideal base nozzle and a nozzle extension with favorable wall pressure gradient (TICNP). The numerical tests was solved on a 100×52 stretched grid. The initial configuration has been arbitrarily chosen as the supersonic flowfield in a costant duct, with the mach number linearly varying from $M_{in} = 1$ to $M_{ou} = 3$. The pressure distribution imposed on the upper boundary is a spline fitting of the experimental data given in Ref.11 and corresponding the case of adapted nozzle. A sequence of the flowfield evolution during the transient is presented in Figure 5. Pressure contour lines are

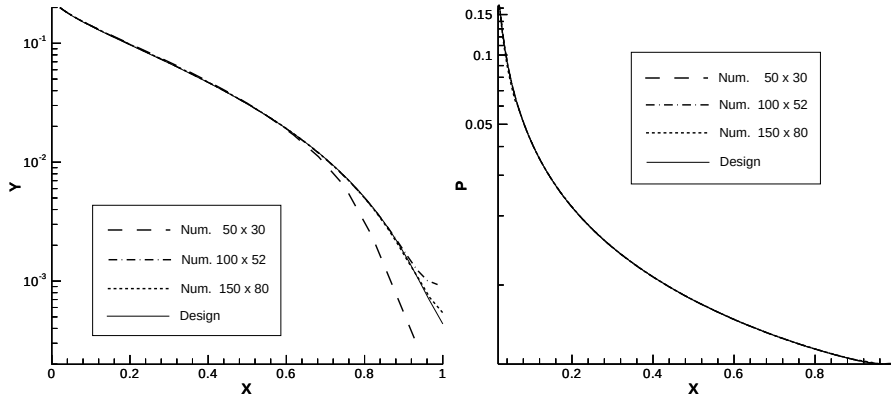


Figure 4. Plug Nozzle. Plug geometry (*left*) and pressure distribution (*right*) comparisons for different grids

also showed in the same figure. Wall pressure data p_w/p_c computed by the inverse method (**cpf**) (dot line) are compared with the experimental results (**epf**) (right triangles) are shown in Figure 6. Since the exact geometry of nozzle of Ref 11 was unknown to us, the geometry has been fixed and some off-design conditions of the nozzle are investigated by a direct solver, putting on evidence that the obtained geometry is the correct one. In fact we observe that the pressure distribution predicted for the computed geometry are in good agreement with the experimental data, as shown in Figure 6. The Pressure data are ranging between $30 \leq p_c/p_a \leq 50$ where p_c and p_a are the chamber and ambient pressure, respectively.

C. Bell Optimization

This example refers to the coupling of the inverse procedure and a gradient based optimization. A classical bell is designed following the guidelines proposed in section IV . The pressure ratio is $pw_{1n}/pw_{ou} = 35$ at wall. The Reynolds number is 10^5, based on the bell length. The initial wall pressure distribution is parabolic, as shown in Fig.10 while the final distribution is the result of the optimization. The inlet flow is sonic and uniform. The initial bell configuration has been computed by solving the viscous inverse problem on a 50×40 stretched grid. The bell shape and the related flowfield are shown in Figure 9a . The initial choice of the wall pressure was intentionally very poor, resulting in a nozzle with bad performances. An approximation of the gradient dT/dm_j is evaluated numerically, solving N inverse problems, one for each wall boundary cell, e.g. $N = 50$. This evaluation is based on the Euler equation, instead of Navier-Stokes equation, to alleviate the computational cost of the procedure. The wall pressure distribution is then updated according to eqs.(13,16) and, again, a viscous inverse problem is solved to find the new nozzle configuration that should realize an higher thrust, while pw_{in} and pw_{ou} remain fixed and the criterion to avoid premature separation is satisfied. The iteration continues until a satisfactory level of optimization is obtained. In the proposed test 24 iteration was enough to decrease the gradient L^2-norm residual of two order of magnitude and to obtain about 400% gain in thrust, as visible in Figs.7-8. Altough this design enhancement is due to the very bad performance of the initial geometry the procedure has shown to be successful and to drive the system in the right direction, even if the starting point was very far from optimality. As a drawback the computation cost required is about a day of CPU on a CompaQ DS-20 workstation. Better performances are expected by replacing the evaluating the functional gradient by adjoint methods[12] .

VI. Conclusions

A viscous, inverse solver based on Navier-Stokes equations has been proposed and validated through some nozzle design experiments. The approach, which essentially tracks a unsteady flow surface until a steady

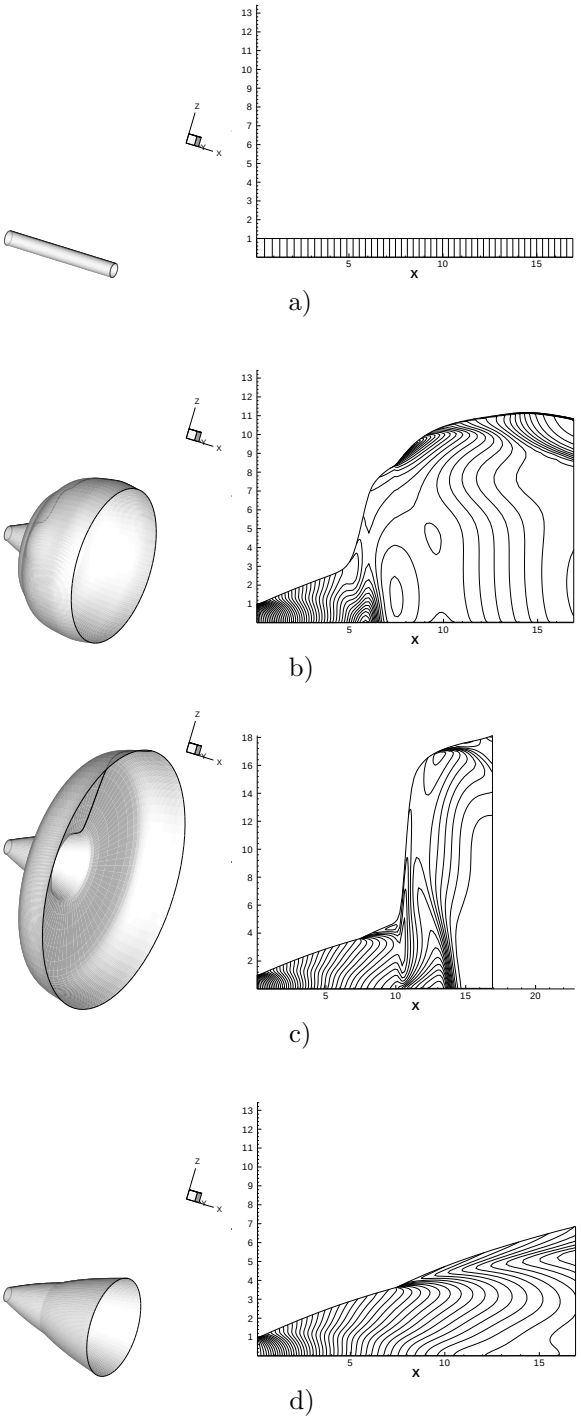


Figure 5. Dual-Bell. Geometry and flowfield evolution: (a) initial; (b,c) transients (d) final configuration.

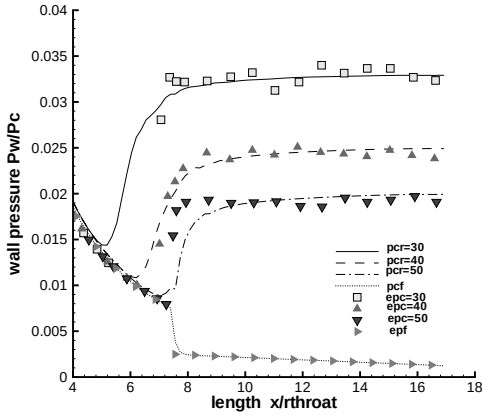


Figure 6. Dual Bell. Wall pressure comparison.

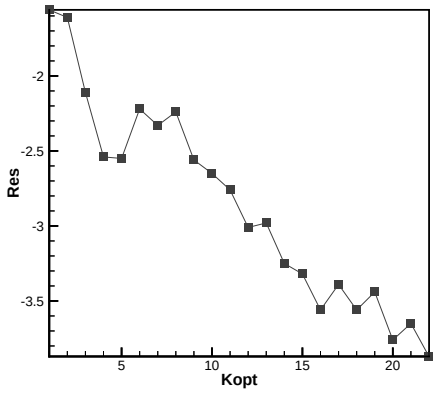


Figure 7. Bell Optimization. Gradient residual versus optimization steps.

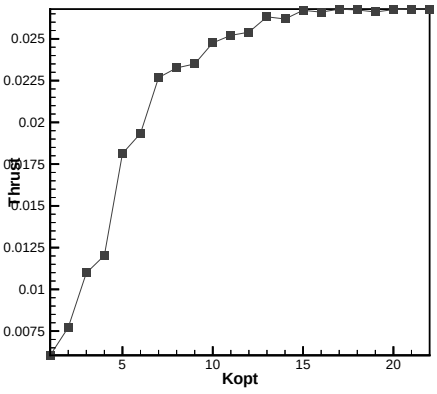


Figure 8. Bell Optimization. Thrust versus optimization steps.

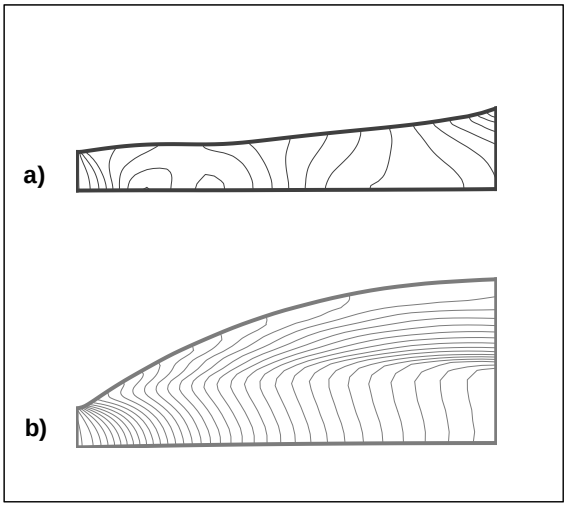


Figure 9. Bell optimization: (a) initial and (b) final bell geometry and pressure field.

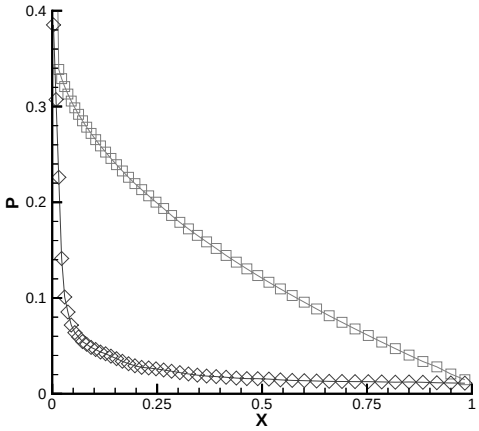


Figure 10. Bell optimization. (−□−) initial and (−◇−) final wall pressure distributions.

configuration is reached, extends to viscous flows a well established procedure for Euler equations. The additional boundary conditions, required for a correct boundary treatment in viscous flows, are introduced in the original technique in the same way as for the "direct problem" solution. The tests have shown that the technique gives an accurate representation of the target geometry and of the flow solution. Since only the steady configuration is of interest, the procedure could be greatly enhanced using over-relaxing techniques. The feasibility of an tool of optimal design that uses the inverse technique coupled to a gradient optimization method is demonstrated. Nevertheless the number of control variables involved in and the related computational cost in evaluating the gradient suggests the use of adjoint methods to gain effectiveness.

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